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# Adaptive Path Following for a Differential Drive Robot with EKF-based Localization

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**Abstract:** For a mobile robot, uncertainties in its operating environment and system setup are common challenges. The robot may need to follow a new path in order to avoid collision with a previously unknown object in the environment. Additionally, the robot does not usually have precise knowledge of its own pose. To compensate for these uncertainties, we develop an adaptive path following scheme for a differential drive robot with localization capabilities. We employ a least-squares adaptive law to estimate an unknown circular path. With measurements to landmarks in the environment, we use an extended Kalman filter to estimate the robot's pose. Furthermore, a cascaded control scheme is used to track the estimated path. Simulation and experimental results are provided to verify the developed adaptive path following scheme and demonstrate its capabilities.

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*Keywords:* Path following, adaptive control, mobile robots, localization.

## 1. INTRODUCTION

In many robotic applications, a robot does not have all information about the environment and its own state. It often relies on different techniques to estimate this information in order to perform its tasks. In this paper, we address path uncertainties in the form of deviations from a pre-planned path to new and previously unknown paths or absence of a pre-planned path. The first form arises, e.g., when the robot's operating environment is dynamic or the robot does not have perfect knowledge of the environment and is subject to possible collision with unknown obstacles (emerging) on its pre-planned path (Phillips and Likhachev (2011)). In such cases, the robot needs to follow a new path based on the geometry and location of the obstacle to avoid collision (Zhang et al. (2016)). An example of the second form is the path uncertainties of a robot moving around to inspect unknown objects (Khan et al. (2021)).

Adaptive control is widely used to allow mobile robots to operate with uncertainties. Uncertainties originating from modeling and disturbances have been widely studied in previous works. Parameter projection is used in Do et al. (2002) to account for modeling uncertainties and environmental disturbances in a sea vessel. Kim and Bentsman (2006) studies disturbance rejection in adaptive control of systems modeled by parabolic or hyperbolic partial differential equations with unknown and varying parameters. Soetanto et al. (2003) estimates uncertain physical parameters of a unicycle robot online for path following tasks. Alves et al. (2018) utilizes a least-squares (LS) adaptive law to identify uncertain model parameters of a robot's DC motor. Through an inverse Lyapunov analysis, Jleilaty et al. (2019) develops a model reference adaptive control scheme to compensate sudden addition of unknown load on mobile robots. There exist further adaptive control designs for compensating system dynamics uncertainties in mobile robot systems in the literature (Zhao et al. (2019)).

Another challenging identification problem in adaptive control of mobile robots, in addition to online identification of uncertain dynamic system parameters, is online estimation of unknown path parameters. To this point, line fitting has been used to approximate unknown initial pose to target paths in Rajurkar et al. (2016), and a gradient-based adaptive path following scheme is designed for mobile robot manipulator platforms in Perez et al. (2020), based on estimation of the path parameters iteratively online utilizing sensor measurements. In this work, we explore an LS based adaptive law to improve the performance of the path estimator. Additionally, we design a cascade motion controller for the robot to follow the estimated path.

In addition to uncertainty in path parameters, uncertain knowledge of the robot's exact pose may impose difficulties in accurate position control. To address this problem, the robot can rely on different sensor information to estimate its pose. Extended Kalman filter (EKF) is a powerful tool for this task (Simanek et al. (2015)). In our work, we use distance measurements to a set of landmarks in the environment to estimate the robot pose with EKF. The proposed adaptive control scheme that is composed of this EKF-based pose estimator, the LS-based path estimator and the cascade motion controller is verified in both simulation tests and physical experiments.

Compared to previous work, including Perez et al. (2020), the main contributions of this paper are: (i) design of an LS-based path parameter estimator (with better performance), (ii) integration of EKF-based localization to adaptive path following, (iii) development of a cascade motion controller to follow the estimated path, and (iv) real-time experimentation on a mobile robot set-up.

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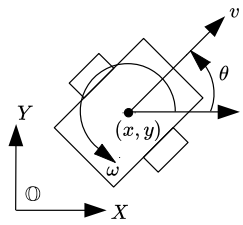


Fig. 1. Schematic of a differential drive robot.

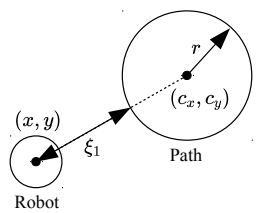


Fig. 2. Illustration of path measurement $\xi_1(t)$.

## 2. PROBLEM FORMULATION

### 2.1 Robot Model

Fig. 1 shows the differential drive robot schematic considered in this work. The robot's motion is constrained in a 2D reference frame $\mathbb{O}$, while its pose at time $t$ is described by the center position $[x(t)\ y(t)]^T$ in $\mathbb{O}$ and orientation angle $\theta(t)$. Under the assumption of negligible friction between the robot and the floor, the robot is governed by the kinematic model

$$\begin{bmatrix}\dot{x}(t)\\ \dot{y}(t)\\ \dot{\theta}(t)\end{bmatrix} = \begin{bmatrix}v(t)\cos\theta(t)\\ v(t)\sin\theta(t)\\ \omega(t)\end{bmatrix} = \begin{bmatrix}v_x(t)\\ v_y(t)\\ \omega(t)\end{bmatrix} = \begin{bmatrix}V(t)\\ \omega(t)\end{bmatrix}, \quad (1)$$

where $V(t) = [v_x(t)\ v_y(t)]^T$, $v(t)$ and $\omega(t)$ are the vectoral velocity, linear speed and angular velocity of the robot, respectively.

### 2.2 Path Following Problem

We consider the motion control problem of having a differential drive robot with kinematics (1) track an uncertain path, the distance $\xi_1$ to which the robot can measure. To simplify the problem, we consider a circular path with unknown radius $r$ and unknown center location $c = [c_x\ c_y]^T$. The distance

$$\xi_1(t) = \left\| \begin{bmatrix}x(t)\\ y(t)\end{bmatrix} - c \right\| - r \quad (2)$$

of the robot's center to the path at time $t$, as illustrated in Fig. 2, and its time derivative

$$\xi_2(t) = \dot{\xi}_1(t). \quad (3)$$

are assumed to be available for measurement, similarly to Perez et al. (2020). In practice, the distance $\xi_1$ may be obtained by a depth camera or a laser scanner, and the time derivative $\xi_2$ can be calculated by numerically differentiating $\xi_1$ and low-pass filtering.

In order for the robot to follow this path, we need to design an algorithm to obtain the center estimate $\hat{c}(t) = [\hat{c}_x(t)\ \hat{c}_y(t)]^T$ and radius estimate $\hat{r}(t)$ at time $t$.

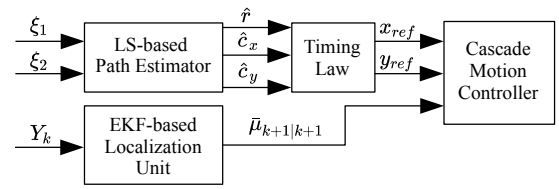


Fig. 3. Overall structure of the control scheme.

### 2.3 Localization

Localization plays an important role in autonomous navigation for a mobile robot. However, localizing the robot by means of odometry only is inaccurate due to the odometry and sensor model uncertainties and sensor noises (Teslić et al. (2011)). The accumulation of these uncertainties and noises over time cause decrease in localization accuracy significantly. To mitigate this effect, we use EKF to utilize its residing advantages (Pomárico-Franquiz et al. (2014)), such as ability to perform denoising for nonlinear systems, low memory usage, and fast processing rate.

## 3. PROPOSED APPROACH

Fig. 3 shows the overall structure of our proposed scheme. An LS-based path estimator utilizes the measured path distance and distance change rate information to estimate the path parameters online. The path parameter estimates are used to generate the reference location trajectory of the robot. Meanwhile, an EKF-based localization unit estimates the robot's pose based on sensor measurements. Both the reference location and pose estimate are sent into a cascade motion controller to move the robot.

The detailed design of the path estimator is presented in Sec. 4. Sec. 5 provides the trajectory generation and tracking control design and Sec. 6 presents the localization unit.

## 4. LS-BASED ONLINE PATH ESTIMATOR DESIGN

Similarly to Perez et al. (2020), in parametric modeling for the estimator design, it is assumed that the robot pose $[x(t)\ y(t)\ \theta(t)]^T$ and its time derivative are known. With this assumption, which is later relaxed by using an EKF-based localization unit, equations (2) and (3) lead to

$$\frac{d}{dt}(\xi_1(t)+r)^2 = \frac{d}{dt}\left(\left(\begin{bmatrix}x(t)\\ y(t)\end{bmatrix} - c\right)^T \left(\begin{bmatrix}x(t)\\ y(t)\end{bmatrix} - c\right)\right),$$

which can be rewritten in linear parametric form as

$$z(t) = \Theta^{*T}\phi(t), \quad (4)$$

where

$$z(t) = [x(t)\ y(t)]\begin{bmatrix}\dot{x}(t)\\ \dot{y}(t)\end{bmatrix} - \xi_1(t)\xi_2(t),$$

$$\phi(t) = [\xi_2(t)\ \dot{x}(t)\ \dot{y}(t)]^T,$$

$$\Theta^* = [r\ c_x\ c_y]^T.$$

The estimation model corresponding to the parametric model in (4), with path parameter estimates $\hat{\Theta}(t) = [\hat{r}(t)\ \hat{c}_x(t)\ \hat{c}_y(t)]^T$, is

$$\hat{z}(t) = \hat{\Theta}^T(t)\phi(t). \quad (5)$$

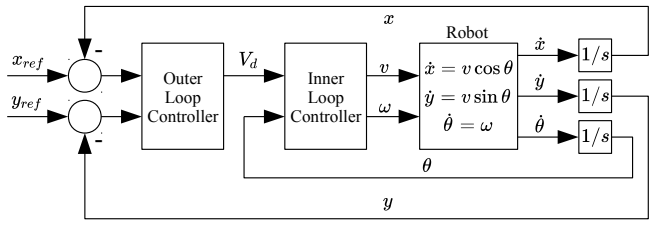


Fig. 4. Controller structure.

Based on the measured parametric model output $z(t)$ and the calculated estimation model output $\hat{z}(t)$, an estimation error is defined as

$$\varepsilon(t) = \frac{z(t) - \hat{z}(t)}{m_s^2(t)}, \tag{6}$$

where $m_s^2(t) = 1 + \alpha\phi(t)^T\phi(t),\ \alpha \geq 0$, is a normalization signal. Instead of the gradient based parameter identifier in Perez et al. (2020), to generate the path parameter estimate $\hat{\Theta}(t)$ with initial value set at $\hat{\Theta}_0 \in \mathbb{R}^3$, the following LS-based online parameter identifier with a forgetting factor (Ioannou and Fidan (2006)) is used;

$$\dot{\hat{\Theta}}(t) = P(t)\varepsilon(t)\phi(t),\ \hat{\Theta}(0) = \hat{\Theta}_0, \tag{7}$$

$$\dot{P}(t) = \beta P(t) - P(t)\frac{\phi(t)\phi^T(t)}{m_s^2(t)}P(t),\ P(0) = P_0 > 0, \tag{8}$$

where $\beta \geq 0$ is the forgetting factor, $P(t) \in \mathbb{R}^{3\times3}$ is the covariance gain matrix, $P_0 \in \mathbb{R}^{3\times3}$ is the initial covariance.

## 5. PATH FOLLOWING MOTION CONTROL DESIGN

The radius estimate $\hat{r}(t)$ and center estimate $\hat{c}(t) = [\hat{c}_x(t)\ \hat{c}_y(t)]^T$ generated by the path estimator fully define the estimated path, along which, the desired location of the robot $[x_{ref}(t)\ y_{ref}(t)]^T$ at time $t$ is expressed by a timing law with arbitrary (design parameter) frequency $\omega_r$ as

$$x_{ref}(t) = \hat{r}(t)\cos\omega_r t + \hat{c}_x(t), \tag{9}$$

$$y_{ref}(t) = \hat{r}(t)\sin\omega_r t + \hat{c}_y(t). \tag{10}$$

Similar to the path estimator design, we derive the motion control law, treating the robot pose known all the time. In the final design, the location estimate output of the EKF-based localization unit will substitute the unknown (uncertain) location. We use a cascade control approach as illustrated in Fig. 4. The outer loop controller generates the reference location trajectory for the robot location online, aiming to minimize the path tracking error. The inner loop controller uses the trajectory generated by the outer loop controller to compute the linear speed $v(t)$ and angular velocity $\omega(t)$ for the robot's motion model in (1).

### 5.1 Outer Loop

From the perspective of the outer loop controller shown in Fig. 4, the plant is essentially an integrator which computes the robot's position $[x(t)\ y(t)]^T$ from its velocity $[v_x(t)\ v_y(t)]^T$. The purpose of the outer loop is to generate the robot's desired velocity $V_d(t) = [v_{dx}(t)\ v_{dy}(t)]^T$ for the inner loop to track. The desired velocity is computed by two proportional-integral (PI) controllers in the form of

$$V_d = \begin{bmatrix} v_{dx} \\ v_{dy} \end{bmatrix} = \begin{bmatrix} \left(K_{p,x} + \dfrac{K_{i,x}}{s}\right)e_x \\ \left(K_{p,y} + \dfrac{K_{i,y}}{s}\right)e_y \end{bmatrix}, \tag{11}$$

where $K_{p,x}$, $K_{p,y}$, $K_{i,x}$, $K_{i,y}$ are the proportional and integral control gains, and $e_x(t) = x_{ref}(t) - x(t)$, $e_y(t) = y_{ref}(t) - y(t)$ are the reference trajectory tracking errors.

### 5.2 Inner Loop

The goal of the inner loop controller is to compute $v(t)$ and $\omega(t)$ such that $v_x(t)$ and $v_y(t)$ in (1) track their desired values $v_{dx}(t)$ and $v_{dy}(t)$, respectively. The desired orientation angle $\theta_d(t)$ for the kinematics (1) is defined to be aligned with $V_d(t)$, i.e.,

$$\theta_d(t) = \text{atan2}(v_{dy}(t), v_{dx}(t)), \tag{12}$$

where $\text{atan2}(y, x)$ denotes the 4-quadrant arctangent value for $(x, y)$, viz. the counterclockwise angle from the positive $x$ axis to the ray to the vector $[x, y]^T$. Based on (12), the linear speed $v(t)$ is generated to satisfy

$$v_{dx}(t) = v(t)\cos\theta_d(t), \tag{13}$$

$$v_{dy}(t) = v(t)\sin\theta_d(t). \tag{14}$$

Hence, we set

$$v(t) = \sqrt{v_{dx}^2(t) + v_{dy}^2(t)}. \tag{15}$$

$\omega(t)$ is generated by the first order lag compensator

$$\omega = C_i(s)e_\theta, \tag{16}$$

with transfer function $C_i(s)$, to drive the orientation tracking error $e_\theta(t) = \theta_d(t) - \theta(t)$ to zero. The zero and pole locations of $C_i(s)$ are selected based on a closed loop pole placement procedure so that the inner loop controller's transient response is faster than that of the outer loop controller to make tracking of $V_d(t)$ feasible.

## 6. EKF-BASED LOCALIZATION

As different from Perez et al. (2020), we further discuss path following with uncertain robot pose information. To estimate the unknown location and heading angle of the robot during motion, using image-based relative position (or equivalently distance and bearing) measurements from a set of landmarks with known constant positions, we employ an EKF-based localization module. In design of the EKF-based localization algorithm, the following discrete-time approximation of the kinematic model (1) is used:

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \\ \theta_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix} + \begin{bmatrix} v_k\cos\theta_k T_s \\ v_k\sin\theta_k T_s \\ \theta_k\omega_k T_s \end{bmatrix}, \tag{17}$$

where $T_s$ is the sampling period and the subscript $k$ for any singal indicates the sampled value of that signal at time instant $t = kT_s$. In practice, (17) is not accurate due to inherited uncertainties such as the slip of the wheels (Hoang et al. (2012)). EKF-based localization can effectively compensate for these uncertainties. Denoting the control signal and the sate vector at time $t = kT_s$ as $u_k = [v_k\ \theta_k]^T$ and $\mu_k = [x_k\ y_k\ \theta_k]$, respectively, a more accurate motion model is obtained as

$$\mu_{k+1} = g(\mu_k, u_k) + n_k = \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix} + \begin{bmatrix} v_k\cos\theta_k T_s \\ v_k\sin\theta_k T_s \\ \omega_k T_s \end{bmatrix} + n_k, \tag{18}$$

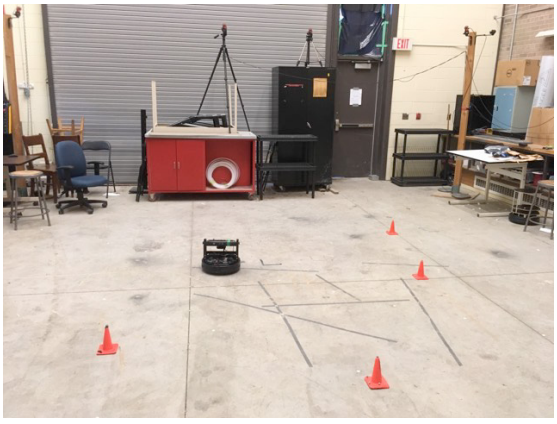

Fig. 5. Experimental setup.

where $n_k \in \mathbb{R}^3$ is a zero-mean Gaussian process noise term that represents the modeling uncertainties.

The robot is assumed to be equipped with a camera or another distance and bearing measurement sensor to obtain information of its surrounding environment. Assume there are $p$ stationary landmarks in the environment, and the robot can measure its distance and bearing angle to these landmarks. For a landmark $i$ located at $(m_{i,x}, m_{i,y})$ in frame $\mathbb{O}$ of Fig. 1, the measurement at time $t = kT_s$ can be modeled as (Thrun et al. (2006))

$$y_{i,k} = h_i(\mu_k) + \eta_i = \begin{bmatrix} \sqrt{(m_{i,x} - x_k)^2 + (m_{i,y} - y_k)^2} \\ \operatorname{atan2}\left(\dfrac{m_{i,y} - y_k}{m_{i,x} - x_k}\right) - \theta_k \end{bmatrix} + \eta_i, \tag{19}$$

where $\eta_i \in \mathbb{R}^2$ denotes the measurement noise that is assumed to follow the Gaussian distribution. Considering all the $p$ landmarks, the complete measurement model is

$$Y_k = h(\mu_k) + \eta = \begin{bmatrix} h_1(\mu_k) \\ \vdots \\ h_p(\mu_k) \end{bmatrix} + \begin{bmatrix} \eta_1 \\ \vdots \\ \eta_p \end{bmatrix}, \tag{20}$$

where $\eta \in \mathbb{R}^{2p}$ captures the measurement noises for all the $p$ landmarks.

The prediction and correction stages of the EKF localization algorithm are set as follows. At each discrete time step $k$, the latest control signal $u_k$ and a posteriori estimate $\hat{\mu}_{k|k}$ of $\mu_k$ are utilized to set the prediction stage as follows:

$$\hat{\mu}_{k+1|k} = g(\hat{\mu}_{k|k}, u_k), \tag{21}$$

$$\Sigma_{k+1|k} = G_{\mu,k}\Sigma_{k|k}G_{\mu,k}^T + \Sigma_n, \tag{22}$$

where $\hat{\mu}_{k+1|k}$ denotes a priori estimate (prediction) of $\mu_{k+1}$, $G_{\mu,k} = \left.\frac{\partial g}{\partial \mu}\right|_{\mu=\hat{\mu}_{k|k}}$, $\Sigma_n$ is the covariance matrix of the process noise $n$, $\Sigma_{k|k}$ is the error covariance estimate from the previous step, and $\Sigma_{k+1|k}$ is the error covariance prediction according to $\Sigma_{k|k}$.

Once the priori state estimate $\hat{\mu}_{k+1|k}$ and the sensor measurement at time $k+1$ are available, the EKF algorithm proceeds to the following update stage:

$$K_k = \Sigma_{k+1|k}H_{\mu,k}^T\left(H_{\mu,k}\Sigma_{k+1|k}H_{\mu,k}^T + \Sigma_\eta\right)^{-1}, \tag{23}$$

$$\hat{\mu}_{k+1|k+1} = \hat{\mu}_{k+1|k} + K_k\left(Y_{k+1} - h\left(\hat{\mu}_{k+1|k}\right)\right), \tag{24}$$

$$\Sigma_{k+1|k+1} = \left(I - K_kH_{\mu,k}\right)\Sigma_{k+1|k}, \tag{25}$$

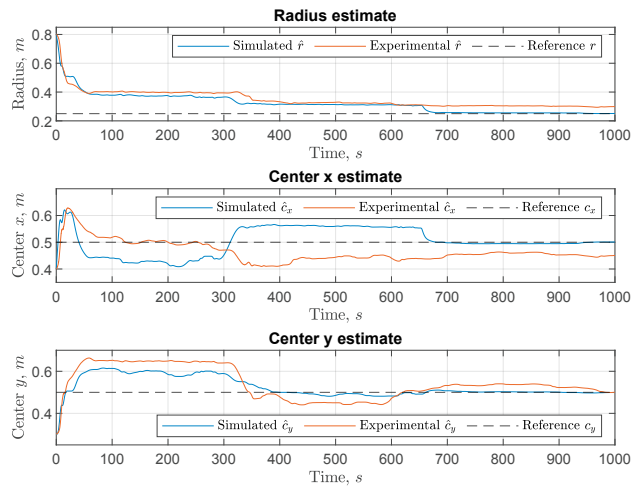


Fig. 6. Path parameter estimates with the gradient adaptive law (Perez et al. (2020)).

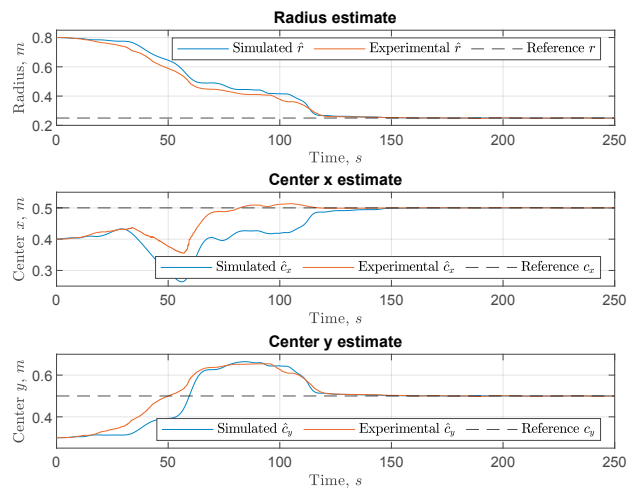


Fig. 7. Path parameter estimates with the LS-based adaptive law (7), (8).

where $K_k$ is the Kalman gain, $H_{\mu,k} = \left.\frac{\partial h}{\partial \mu}\right|_{\mu=\hat{\mu}_{k+1|k}}$, $\Sigma_\eta$ is the covariance of the measurement noise $\eta$, $\hat{\mu}_{k+1|k+1}$ and $\hat{\Sigma}_{k+1|k+1}$ are the final state and error covariance estimates given the sensor data $Y_{k+1}$. $\hat{\mu}_{k+1|k+1}$ is used to substitute the unknown pose in the design presented in Sec. 5.

## 7. SIMULATION AND EXPERIMENTAL RESULTS

To verify the scheme proposed in Sections 3-6, we performed simulations and experiments on a physical robot setup composed of a Quanser Qbot 2e, an Optitrack camera tracking system and four landmarks as shown in Fig. 5. The poses of the robot and landmarks are captured by the Optitrack system. (19) is applied to these measured poses to mimic landmark distance and bearing measurements.

### 7.1 Path Estimator

We first evaluate the online path parameter identifier (7), and compare its performance with that of the gradient parameter identifier

$$\dot{\hat{\Theta}}(t) = \Gamma\varepsilon(t)\phi(t) \tag{26}$$

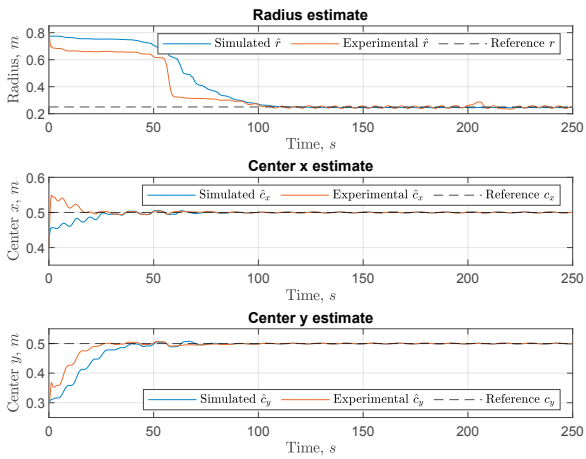


Fig. 8. Path parameter estimates for path following.

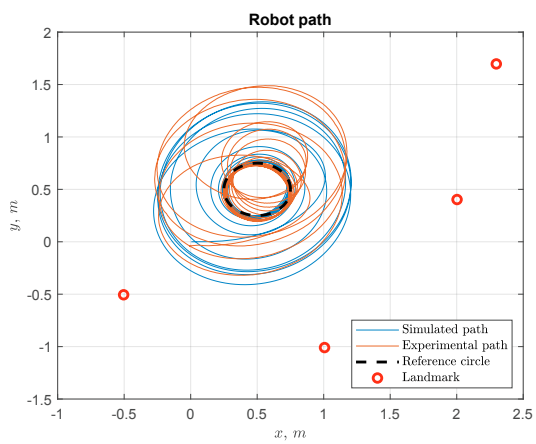


Fig. 9. Robot path obtained by the full system.

used in Perez et al. (2020), where $\Gamma$ is a constant positive definite symmetric $3 \times 3$ gain matrix. In both configurations, the path to be estimated is defined as a circle with its center at $[c_x \ c_y]^T = [0.5 \ 0.5]^T$ m and with a radius of $r = 0.25$ m. The initial path estimate is chosen as $\hat{\Theta}_0 = [\hat{r}_0 \ \hat{c}_{x0} \ \hat{c}_{y0}]^T = [0.8 \ 0.4 \ 0.3]^T$ m. The robot is actuated with $v = 0.1$ m/s and $\omega = 0.2 \sin 0.08t + 0.1 \sin 0.2t + 0.2 \sin 0.15t$ rad/s. The normalization constant $\alpha$ is set to 0.005. The adaptive gain for the gradient-based adaptive law is set as $\Gamma = 80 I_{3\times3}$. In the LS algorithm, the forgetting factor is set as $\beta = 0.1$ and the initial error covariance is $P_0 = I_{3\times3}$. $\omega_r = 0.5$ is used in (9), (10).

Fig. 6 shows the path parameter estimates using the gradient-based adaptive law. In simulation, all estimated parameters reach the correct values at steady state after about 700 seconds. The experimental results for the radius estimate $\hat{r}$ and the path center estimate $\hat{c}_y$ follow similar trends as the simulation results. However, the experimental $\hat{c}_x$ deviates from the simulated $\hat{c}_x$ significantly. Moreover, none of the experimental estimates can reach the correct values after the experiment runs for 1000 seconds.

Fig. 7 shows the results obtained by the LS algorithm with a forgetting factor. The results recorded from the experiment and simulation resemble each other both in terms of transient and steady state performance. All three parameter estimates reach steady state at their

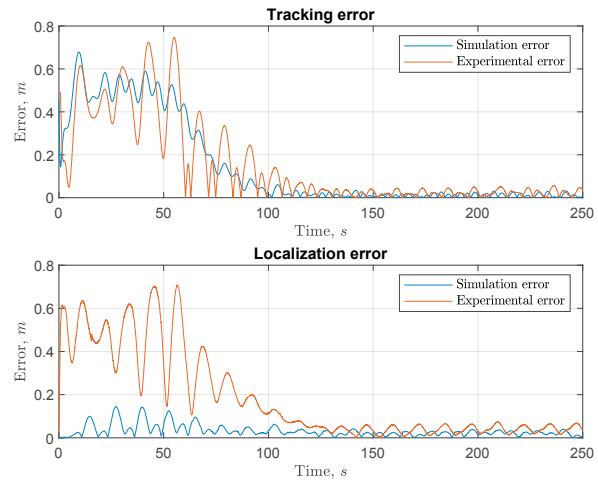


Fig. 10. Tracking error and localization error based on the full system.

corresponding reference values after the simulation and experiment run for 150 seconds.

From the results above, it can be seen that the LS-based adaptive law with a forgetting factor can provide more accurate path parameter estimates with only about 20% execution time as the gradient-based law does.

### 7.2 Path Following with EKF-based Localization

Using the LS-based online path parameter identifier (7), (8), the cascade controller (9)-(16), and EKF-based localization scheme (21)-(25), we have conducted a set of simulations and experiments to verify our adaptive path tracking control design and demonstrate its performance. The path parameters and path estimator settings are identical to the ones in Sec. 7.1. We choose the outer loop controller gains as $K_{p,x} = 1$, $K_{i,x} = 0.005$, $K_{p,x} = 1.1$, $K_{i,x} = 0.0055$. The inner loop controller is set as

$$C_i(s) = \frac{s + 36}{s + 11}, \tag{27}$$

which leads to two repeated poles at $s = -6$ for the inner loop. These two poles are significantly faster than the outer loop poles, which are approximately located at $s = -1$ and $s = -0.005$ for both $C_{o,x}$ and $C_{o,y}$. The robot is placed at an initial pose close to $[0 \ 0 \ 0]^T$.

For facilitating the EKF-based localization scheme, we consider four landmarks in the robot's operating environment with locations at $(2.0, 0.4)$, $(1.0, -1.0)$, $(2.3, 1.7)$, and $(-0.5, -0.5)$, all in meters, with respect to frame $\mathbb{O}$. The process noise $n$ is assumed to follow a normal distribution $\mathcal{N} \sim (0, 10^{-7})$ and the measurement noise $\eta$ is assumed to be $\mathcal{N} \sim (0, 0.1)$. These assumptions and landmark setup result in the process noise covariance as $\Sigma_n = 10^{-7} \cdot I_{3\times3}$ and the measurement noise covariance as $\Sigma_\eta = 0.1 \cdot I_{8\times8}$. The localization is initialized with $\hat{\mu}_{0|0} = [0 \ 0 \ 0]^T$ and $\Sigma_{0|0} = 0_{3\times3}$. The sampling rate is $T_s = 0.05$ sec.

The simulation and experimental results are shown in Figures 8-10. Fig. 8 shows the path parameter estimates after the controller and the EKF-based localization unit are used. In both simulation and experiment, the estimated

path parameters converge to the reference values. Compared to the case in Sec. 7.1 where the control inputs are predefined, the path estimates in adaptive path following reach steady state faster.

Fig. 9 shows the robot's actual location trajectory on $x$-$y$ plane in both simulation and experiment, landmark positions as well as the reference path. Fig. 10 shows the tracking error and localization error, which are defined by

$$E_{track}(k) = \left| \sqrt{(x_k - c_x)^2 + (y_k - c_y)^2} - r \right|, \quad (28)$$

$$E_{EKF}(k) = \sqrt{(x_k - \hat{x}_k)^2 + (y_k - \hat{y}_k)^2}, \quad (29)$$

where $\hat{\mu}_{k|k} = \begin{bmatrix} \hat{x}_k & \hat{y}_k & \hat{\theta}_k \end{bmatrix}^T$.

Based on the results, the robot initially follows a circular path with larger error. As the path parameter estimates become more accurate, the robot begins to follow the correct path. As indicated in Fig. 10, the robot's trajectory converges to close vicinity of the reference circle after $t = 100$ sec in simulation and experiment. Compared to the simulation, the tracking error in our experiment is larger. This is because transient effects are not considered in simulation while they are inevitable in a physical system.

The localization errors for both cases also converge to approximately 0. One significant discrepancy between the experimental and simulation result is that the experimental localization error for the first 150 seconds is considerably larger. One cause for this discrepancy is the robot's initial localization error. In simulation, the robot starts from an initial pose at $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T$, resulting in zero initial error. While in the experiment, the robot is initially placed at a pose close to $\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T$, leading to nonzero initial error. Another possible causes is that the chosen process noise and measurement noise are different from the actual noises on the physical robot and camera system.

## 8. CONCLUSION AND FUTURE WORK

In this work, we have proposed a framework for a mobile robot to estimate the parameters of an unknown circular path online and to follow this path under localization uncertainty. The framework utilizes a fast and accurate LS based online parameter identification algorithm with a forgetting factor to generate path parameter estimates. Then distances and bearings to a set of landmarks are processed by an EKF based localization scheme to provide robot location and heading angle estimates. The online path parameter identifier and the EKF based localization scheme are integrated with a two-stage cascade controller to generate the linear and angular robot velocities for tracking the unknown reference path.

As follow-up studies, it is planned to investigate other more advanced controllers to improve the path tracking performance. This includes exploration of direct adaptive control designs for the same path tracking problem. Extending the framework to other (arbitrary) path geometries is another potential future research direction.

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